🧠 Neurophysiotherapy OSCE Prep

Cases 7 & 8 — Clinical Reasoning Guide

Target: Very Good (70%+) | Physiotherapist Perspective Throughout

Case 7: Cerebellar Ataxia · Subjective · Pathophysiology · Differentials · Objective · Treatment · Case 8: FND · Subjective · Pathophysiology · Differentials · Objective · Treatment

Case 7 — Ms. Anika Gupta

41 F · Slow-onset Cerebellar Ataxia · Unknown Aetiology

Wide-based gait · Dysmetria · Dysdiadochokinesia · Dysarthria · Fine motor impairment · Preserved strength

1 Subjective Assessment

CONSENT / OPENING STATEMENT

> "Hello Anika, my name is [Name] and I'm one of the physiotherapists here today. Before we start I want to make sure you're comfortable — we'll begin with a conversation about what's been happening, and then I'll do some physical tests. Everything stays confidential within the team, and you can pause or stop at any time. Do you consent to proceeding?"
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> *I establish consent first because it is an ethical non-negotiable, and it also sets a collaborative, respectful tone — important in progressive neurological conditions where patients often feel they are losing control of their own bodies.*

KEY QUESTIONS + CLINICAL REASONING

"Can you describe when you first noticed these symptoms, and whether they have been getting gradually worse, staying the same, or fluctuating?"

> *Establishes the timeline and trajectory. Slow progressive onset over months to years points toward a degenerative or hereditary cause (e.g., spinocerebellar ataxia), while a sudden onset would raise concern for a vascular event, demyelination, or toxic cause. Trajectory directly shapes my urgency of referral and long-term prognosis discussion.*

"Does anyone in your family have similar symptoms — unsteady walking, a tremor, or a diagnosed neurological condition?"

> *Hereditary spinocerebellar ataxias (SCA types 1, 2, 3, 6) are autosomal dominant — a positive family history dramatically increases this probability and would trigger a genetics referral. Even a negative history is informative as it raises the likelihood of acquired causes.*

"Have you had any changes to your vision, any difficulty swallowing, or noticed changes in your memory or concentration?"

> *Brainstem connections to the cerebellum can produce nystagmus, diplopia, and dysphagia — their presence widens the diagnostic picture toward posterior fossa pathology or MSA. Cognitive change may indicate a more diffuse neurodegenerative process. Dysphagia is an immediate safety concern: aspiration risk must be communicated to the MDT.*

"Do you drink alcohol regularly, or have you been on any long-term medications such as anti-epileptics like phenytoin?"

> *Chronic alcohol use causes dose-dependent Purkinje cell loss, producing a predominantly gait ataxia picture. Phenytoin toxicity is a reversible cause of cerebellar signs. Identifying a toxic cause is clinically actionable — removal of the agent can halt or partially reverse the condition.*

"Have you had any falls? If so, how often, and do you tend to fall in a particular direction?"

> *Falls are my immediate safety priority as a physiotherapist. Cerebellar ataxia causes multidirectional falls due to truncal instability. Fall frequency informs urgency of gait aid prescription and home safety review. I also use this to baseline the Falls Efficacy Scale.*

"How is all of this affecting your daily life — things like getting dressed, eating meals, getting around, and your social activities?"

> *This patient-centred question directly links physical impairments to activity limitations and participation restrictions. It reveals what matters most to Anika and allows me to build a goal-directed rehabilitation plan. It also opens a conversation about psychosocial impact that I can use to screen for low mood and reduced quality of life.*

RED FLAGS

RED FLAG	CLINICAL SIGNIFICANCE	ACTION IF PRESENT
Sudden acute onset or rapid recent worsening	Suggests cerebellar haemorrhage, posterior fossa infarct, or mass effect — a potential neurosurgical emergency	Do not proceed with physiotherapy — immediate emergency medical referral
Dysphagia or choking on food/liquids	Risk of aspiration pneumonia; indicates brainstem involvement, alters prognosis	Halt session; urgent SLT referral; document and flag to medical team
Severe or thunderclap headache / neck stiffness	Raised intracranial pressure, posterior fossa lesion, meningitis, or subarachnoid haemorrhage	Urgent medical review; do not continue physiotherapy session
Unexplained significant weight loss	May indicate paraneoplastic cerebellar degeneration — immune-mediated Purkinje cell destruction driven by an occult malignancy (e.g., ovarian, lung, breast)	Flag urgently to medical team; oncology/neurology referral needed
Loss of consciousness or seizures	Epilepsy co-exists with some hereditary ataxias; syncope during falls creates severe injury risk	Do not exercise without medical clearance; document and report to team

PROBLEM LIST

PHYSICAL IMPAIRMENTS (BODY STRUCTURE & FUNCTION)

- Dysmetria — limb incoordination on reaching tasks
- Dysdiadochokinesia — impaired rapid alternating movements
- Truncal and gait ataxia — wide-based, unsteady gait
- Cerebellar dysarthria — slurred, scanning speech
- Impaired postural stability and dynamic balance
- Fine motor dysfunction — grip and object manipulation

FUNCTIONAL ACTIVITY LIMITATIONS & PARTICIPATION RESTRICTIONS

- Unable to safely walk on uneven terrain or in crowds
- Difficulty using cutlery — nutrition and independence at risk
- Cannot fasten buttons or do up zips — reduced ADL independence
- Reduced community mobility and social participation
- Potential impact on employment depending on role demands

PSYCHOSOCIAL / YELLOW FLAGS

Progressive neurological conditions of unknown cause generate significant anxiety and existential uncertainty. Anika may be experiencing grief over loss of function and identity. Fear of falling commonly leads to activity avoidance (kinesiophobia), which paradoxically worsens deconditioning and secondary complications. I would screen formally with the Falls Efficacy Scale — International (FES-I) and ask directly about mood and support at home.

2 Pathophysiology of Working Diagnosis

WHAT IS HAPPENING NEUROLOGICALLY

> The cerebellum functions as the brain's real-time error-correction system — it continuously compares the intended motor command (efference copy from motor cortex) with actual movement feedback from vestibular receptors, proprioceptors, and visual input, then sends corrective output via the deep cerebellar nuclei (primarily the dentate nucleus) through the thalamus to the motor cortex. The key output neurons are
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> **PURKINJE CELLS**
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> , which provide the cerebellum's sole inhibitory output to the deep nuclei. In cerebellar ataxia, Purkinje cells are progressively damaged or lost — without their inhibitory influence, motor output from the deep nuclei becomes disorganised, irregular in timing, and inaccurate in force scaling. The result is the signature triad of
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> **DYSMETRIA**
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> (inability to accurately gauge the distance and force of a movement),
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> **DYSDIADOCHOKINESIA**
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> (loss of rhythm in rapid alternating movements), and
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> **INTENTION TREMOR**
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> (oscillation that worsens as the limb approaches its target, reflecting the absence of corrective dampening). The cerebellar vermis and spinocerebellar tracts govern axial and gait coordination — degeneration here produces the wide-based, lurching "drunken" gait. Cerebellar connections to the brainstem also coordinate articulatory muscles, so Purkinje cell loss produces the characteristic scanning dysarthria heard in Anika's speech.

WHY THIS PATIENT FITS THE DIAGNOSIS

Wide-based "drunken" gait
— truncal ataxia from vermis and spinocerebellar tract degeneration; wide stance compensates for postural instability

Positive finger-to-nose dysmetria
— cerebellar hemisphere dysfunction causing limb incoordination and inability to dampen terminal trajectory

Dysdiadochokinesia
— loss of precise timing and rhythm in rapid alternating movements; a hallmark of cerebellar pathology that cannot be explained by weakness or sensory loss

Slurred/scanning speech
— cerebellar dysarthria from disruption of cerebellar-brainstem circuits controlling laryngeal and articulatory muscles

Preserved bilateral strength
— rules out corticospinal (UMN) or peripheral (LMN) damage as the primary driver; the deficit is purely one of motor coordination

Slow progressive onset
— consistent with degenerative Purkinje cell loss rather than acute vascular event or inflammatory demyelination

3 Differential Diagnoses

✦ Primary Diagnosis: Hereditary Spinocerebellar Ataxia (SCA)

Supporting features: Slow progressive bilateral cerebellar syndrome in a middle-aged adult; dysarthria; fine motor and gait involvement; intact strength. SCA types 1, 2, 3 (Machado-Joseph Disease), and 6 are the most prevalent autosomal dominant hereditary ataxias. Genetic testing using trinucleotide repeat panel analysis is the gold standard confirmatory test. Family history, if positive, would make this near-certain.

Differential 1: Alcoholic / Toxic Cerebellar Degeneration

Supporting features: Progressive bilateral cerebellar syndrome in an adult. However, alcoholic ataxia preferentially affects the *anterior and superior vermis*, causing gait ataxia with relatively preserved upper limb coordination. The prominent upper limb dysmetria and dysdiadochokinesia in Anika's case implicates cerebellar hemisphere involvement, making a purely toxic cause less consistent — though alcohol history must be explicitly excluded.

Differential 2: Multiple System Atrophy — Cerebellar Type (MSA-C)

Supporting features: Progressive adult-onset cerebellar ataxia with dysarthria. MSA-C would be supported by additional features: autonomic dysfunction (orthostatic hypotension, bladder incontinence), early postural instability, and Parkinsonian features. The absence of these features in Anika's presentation makes MSA-C less probable, though it remains in the differential given the unknown aetiology.

WHY I FAVOUR HEREDITARY SCA

I favour hereditary SCA over toxic degeneration because the upper limb incoordination (dysmetria, dysdiadochokinesia) and fine motor impairment are prominent — this pattern reflects cerebellar hemisphere involvement beyond the vermis-predominant picture of alcoholic ataxia. I favour SCA over MSA-C because there is no description of autonomic dysfunction or Parkinsonism, and the age of onset is younger than the typical MSA-C presentation (usually 50s+). Genetic panel testing in conjunction with neurology review would confirm the diagnosis.

4 Objective Assessment — Special Tests

SPECIAL TESTS

Test 1 — Finger-to-Nose Test (FNT)

MEASURES Limb dysmetria and intention tremor — integrity of cerebellar hemisphere coordination and sensorimotor calibration

HOW TO PERFORM

1. Seat the patient comfortably. Position yourself opposite, holding your index finger at arm's length from the patient.
2. Ask the patient to touch the tip of their own nose with their index finger, then touch your finger tip, alternating repeatedly at a moderate pace.
3. Reposition your finger to different locations in space between each reach to vary the movement direction and distance.
4. Observe for: overshoot or undershoot of the target (dysmetria), an oscillating tremor that increases as the finger approaches the target (intention tremor), and irregularity of movement velocity.
5. Test both arms and note any asymmetry.

POSITIVE FINDING Consistent overshooting or undershooting and/or worsening tremor approaching the target = dysmetria and intention tremor indicating ipsilateral cerebellar hemisphere dysfunction.

WHY I CHOSE IT Anika's complaint of difficulty using a fork and buttons strongly suggested dysmetria. This is the gold-standard bedside test for cerebellar limb coordination — it directly maps to her functional complaint, is reproducible, and can be re-tested at each session to track change.

Test 2 — Heel-to-Shin Test (HTS)

MEASURES Lower limb cerebellar coordination — tests cerebellar hemisphere and spinocerebellar tract function in the legs

HOW TO PERFORM

1. Ask the patient to lie supine on the plinth.
2. Instruct them to place the heel of one foot precisely on the opposite knee.
3. Ask them to slide the heel smoothly and accurately down the front of the shin toward the ankle.
4. Observe for: deviation of the heel off the shin laterally (ataxia), jerky/irregular speed, or inability to maintain contact with the shin throughout the movement.
5. Repeat on both sides and compare symmetry.

POSITIVE FINDING Heel deviates off the shin, moves in a jerky or wavering manner, or overshoots the knee = lower limb cerebellar ataxia.

WHY I CHOSE IT Anika's gait presentation implies lower limb incoordination. This test lets me determine whether her ataxic gait is driven by cerebellar lower limb dysfunction specifically — informing whether gait retraining should primarily target coordination, balance, or both.

Test 3 — Romberg Test (Eyes Open vs. Closed)

MEASURES Distinguishes cerebellar ataxia (unstable with eyes OPEN) from sensory/dorsal column ataxia (unstable only with eyes CLOSED) — identifies which sensory modality is most contributing to instability

HOW TO PERFORM

1. Ask the patient to stand with feet together, arms at sides or folded across chest.
2. Stand directly beside the patient, close enough to catch them — never leave them unsupported.
3. Observe for 30 seconds with eyes OPEN — document any sway, stepping, or loss of balance.
4. Ask the patient to close their eyes — observe for a further 30 seconds.
5. A positive Romberg = marked increase in sway or loss of balance upon eye closure (indicating sensory ataxia). Instability even with eyes open = cerebellar ataxia.

POSITIVE FINDING (Cerebellar) Unsteady even with eyes open — cerebellar ataxia is NOT correctable by visual input, unlike dorsal column sensory ataxia. A purely positive Romberg (only on eye closure) would shift suspicion to proprioceptive/dorsal column pathology.

WHY I CHOSE IT This test directly guides my treatment strategy. If Anika can partially compensate with visual input (semi-negative Romberg), I can incorporate visual biofeedback strategies in balance training. If she is unstable even with eyes open, my approach must focus on central coordination retraining rather than sensory substitution.

KEY NEUROLOGICAL OBSERVATIONS

Tone:
Typically *hypotonic* in cerebellar pathology — assess passively at the wrist and knee; muscles may feel "floppy" or pendular on passive movement

Coordination:
Observe dysdiadochokinesia via rapid forearm pronation/supination — loss of rhythm and regularity with increasing speed confirms cerebellar involvement

Gait:
Wide-based, irregular stride length, truncal sway, tendency to veer; I also assess tandem (heel-to-toe) walking and turning — both are particularly impaired in cerebellar ataxia

Posture:
Assess sitting balance on plinth versus standing — truncal instability worsens as base of support narrows; also look for titubation (rhythmic head/trunk tremor at rest)

Oculomotor:
Look for nystagmus, particularly gaze-evoked horizontal nystagmus — present in cerebellar or vestibulo-cerebellar dysfunction; relevant for driving and falls risk

Speech:
Dysarthria quality — cerebellar dysarthria is scanning/staccato; consistent with what is described in Anika's presentation

OUTCOME MEASURE

> **PRIMARY: SCALE FOR THE ASSESSMENT AND RATING OF ATAXIA (SARA)**
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> I would use the SARA — a validated, disease-specific 8-item scale covering gait, stance, sitting balance, speech, finger chase, finger-nose-finger, fast alternating hand movements, and heel-shin slide. Scored 0–40 (higher = more severe ataxia). It has established reliability and responsiveness to change in degenerative ataxias, making it ideal for tracking Anika's rehabilitation progress over time.
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> **SECONDARY: BERG BALANCE SCALE (BBS)**
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> — provides a quantified falls risk score (≤ 45/56 = elevated risk) that is widely understood by the MDT and directly relevant to Anika's primary safety concern.

5 Treatment Plan

INITIAL PHASE — WEEKS 1–4 (2–3 SESSIONS/WEEK)

> **NEUROPHYSIO RATIONALE**
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> Cerebellar rehabilitation leverages
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> **ERROR-BASED MOTOR LEARNING**
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> and
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> **NEURAL PLASTICITY**
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> . The cerebellum is fundamentally a learning structure — it modifies synaptic weights in response to motor errors. Even in the context of Purkinje cell loss, the remaining cerebellar circuits and compensatory cortical pathways can adapt through high-repetition, task-specific, progressively challenging practice. I therefore prioritise active coordination training over passive techniques, always pitching the difficulty at the "just right" challenge level where errors occur but task completion remains possible — this is the zone of optimal learning.

Exercise 1 — Seated Target Reaching (Upper Limb Coordination)

What: Patient seated at a table. Multiple coloured stickers or objects placed at varying distances and heights in front of them. Patient touches each target with their index finger in a specified sequence, returning to their nose between each reach — directly mirroring the FNT but as a functional training task.

Why: This task provides the brain with continuous error signals (over- or undershooting of targets) that the remaining cerebellar circuits and compensatory cortical areas use to recalibrate motor output. High repetition drives use-dependent plasticity. Varying target positions prevents habituation and forces adaptive recalibration.

Dosage: 3 sets × 20 repetitions per arm, 90-second rest between sets.

⬇ Regression	⬆ Progression
Large, close targets (within arm's length). Forearm resting on table to reduce proximal demand. Single direction (forward reach only). Slow, deliberate pace.	Smaller targets at greater distances and varying heights. Remove forearm support. Add time pressure. Progress to standing. Introduce dual task (count backwards aloud) to reduce conscious monitoring of the limb.

Exercise 2 — Frenkel's Exercises (Lower Limb Coordination & Gait)

What: Graded lower limb coordination exercises developed specifically for ataxia rehabilitation. Begin supine: slide heel along a marked line, place foot precisely on numbered floor targets. Progress to sitting: foot placement on marked footprints with visual feedback. Then standing and walking: step precisely onto footprint markers on the floor.

Why: Frenkel's exercises use explicit visual and verbal feedback to substitute for impaired cerebellar proprioceptive integration — they train compensatory motor programmes using intact visual and prefrontal pathways. They are directly translatable to Anika's functional gait complaint and have been used in ataxia rehabilitation for over a century with good clinical support.

⬇ Regression	⬆ Progression
Supine position, wide target marks, slow speed, verbal cueing ("now slide... now stop"). Therapist provides hands-on guidance initially for the motor pattern.	Sitting → standing → walking with floor targets. Reduce visual cues progressively. Add tandem walking. Introduce varied surfaces (foam, slight incline). Progress to community environments when outdoor safety is confirmed.

SAFETY CONSIDERATIONS

- Always use a gait belt and have a spotter during all balance and gait activities — Anika can fall multidirectionally
- Non-slip footwear and clear, uncluttered gym space are mandatory
- Home exercise programme: prescribe only seated or supported exercises initially when unsupervised
- Discuss fall management strategy: how to fall safely (guided descent), how to get up from the floor, and suitability of a personal emergency response alarm
- Assess and prescribe appropriate walking aid — a wider base of support (rollator frame, quadstick) reduces fall risk while allowing rehabilitation to progress safely in parallel

PATIENT / CARER EDUCATION — KEY MESSAGES

"The cerebellum is a learning centre of the brain — the more you practise these exercises consistently, the more your nervous system adapts and finds new ways to coordinate movement. Repetition is the medicine here."

"It is normal to make errors during these exercises — in fact, errors are what teach your brain to recalibrate. Struggling a little is a sign you are working in the right zone."

"Falls are one of our biggest concerns, so we are going to make your environment as safe as possible and teach you strategies for if a fall does happen. Please do not try balance exercises without your safety measures at home."

"Staying active within safe limits is really important — reducing all activity to avoid falling can actually make your balance and strength worse over time. We will pace this carefully together."

PROGRESSION CRITERIA — MEASURABLE TARGETS

- ✓ SARA score reduces by ≥ 2 points (established minimum clinically important difference)
- ✓ Berg Balance Scale score improves to ≥ 45/56 (threshold for reduced fall risk category)
- ✓ Anika can walk 10 metres on a flat surface with no more than 2 lateral steps/stumbles using prescribed walking aid
- ✓ Finger-to-nose dysmetria reduces to ≤ 2 cm target error on 4 of 5 consecutive trials
- ✓ Self-reported improvement in ADL tasks on PSFS (e.g., able to use fork with minimal spillage most meals)

LONG-TERM PLAN — WEEKS 4–12+

> Transition to community-based neurological rehabilitation group where available — peer support is particularly motivating in progressive conditions. Introduce evidence-informed activity programmes: dance therapy, Tai Chi, and aquatic physiotherapy all have emerging evidence for balance, coordination, and falls reduction in ataxia populations. MDT liaison: occupational therapy for adaptive equipment (weighted cutlery, button hooks, grab rails, bathroom aids); SLT for dysarthria management and ongoing swallowing monitoring; neurologist for genetic counselling, disease monitoring, and to discuss any disease-modifying clinical trials. Establish a sustainable self-management plan including a home exercise programme, fatigue management strategies (common in degenerative ataxia), and regular physiotherapy review every 8–12 weeks for programme adjustment.

🎓 Examiner Demonstration — Techniques Performable in the Room

1. **Guided Reaching Task:**

   Place two objects on the table at different distances. Physically guide Anika's arm through a slow, accurate reaching movement using the hand-over-hand technique, then progressively fade assistance — narrate this as the "physical guidance" phase of motor learning, used in early cerebellar rehabilitation before the patient can self-correct reliably.

2. **Tandem Stance Challenge:**

   Demonstrate heel-to-toe stance with contact guarding (hand at the patient's shoulder, ready to catch) — explain this progressively challenges the medial cerebellar and vestibular systems and directly maps to the tandem gait progression goal.

END OF CASE 7 — BEGIN CASE 8

Case 8 — Ms. Chloe Davies

27 F · Functional Neurological Disorder (FND) — Motor Subtype

Left leg drag / weakness · Left arm tremor · Inconsistent on testing · Normal MRI & NCS · Precipitant: Psychological stress · Positive Hoover sign

1 Subjective Assessment

CONSENT / OPENING STATEMENT

> "Hello Chloe, my name is [Name] and I'm one of the physiotherapists here today. Thank you for coming — I know this has been a really difficult time. We're going to start with a conversation about what's been going on, and then I'll do some physical assessments. Everything is confidential within the care team, and you are completely in control — you can pause or stop at any time. Does that sound okay?"
>
> *I use a warm, unhurried tone from the very first words. FND is commonly associated with prior traumatic or dismissive medical encounters — patients often arrive having been told "there's nothing wrong with you," which is deeply invalidating and erodes trust. My opening signals that I believe her symptoms are real and that I am here to understand and help, not to challenge her.*

KEY QUESTIONS + CLINICAL REASONING

"Can you tell me in your own words what happened — how did this all start?"

> *Open-ended question that allows Chloe to narrate from her own perspective without me leading or suggesting. In FND, the patient's illness narrative matters clinically — the sudden onset, the circumstances surrounding it, and whether Chloe herself perceives a link between the stress and the symptoms all shape both the assessment and the treatment approach.*

"Do the symptoms fluctuate throughout the day? Are there specific situations, emotions, or activities where they are better or worse?"

> *Variability and context-dependence are hallmarks of FND and are positive diagnostic features — not signs of malingering. Identifying contexts where symptoms improve (distraction, relaxation, enjoyable activity) gives me therapeutic entry points, because these are conditions I can replicate or build on in physiotherapy.*

"Have you been through anything particularly stressful, traumatic, or emotionally significant around the time this started?"

> *Documents the psychosocial precipitant — asked sensitively, not accusatorially. I frame this as: "We know the nervous system is sensitive to stress in many ways — like how stress causes migraines or an upset stomach. This helps us understand the full picture." I do not suggest that symptoms are therefore not real or self-caused.*

"How has this affected your daily life — your work, relationships, getting around, and how you feel about yourself?"

> *Quantifies functional impact and participation restrictions. Also opens a natural conversation about emotional wellbeing — FND has a high comorbidity with anxiety and depression (60–80% of cases). This information is essential for ensuring the MDT (particularly psychology) is involved at the right intensity.*

"Have you seen any other doctors or specialists about this? What did they tell you, and how did that feel?"

> *Explores prior medical encounters and Chloe's beliefs about her diagnosis. Many FND patients have been dismissed or disbelieved, which creates layers of healthcare-related anxiety and distrust. Understanding what she has been told helps me calibrate my explanation and avoid inadvertently repeating messaging that felt invalidating.*

"Do you have a history of any other health conditions, mental health difficulties, or significant past experiences such as trauma?"

> *FND is associated with higher rates of childhood trauma, PTSD, anxiety, and depression — not because it is "all psychological," but because adverse life experiences sensitise the brain's threat and predictive processing systems. This information shapes the MDT composition needed, and alerts me to avoid certain handling approaches that could trigger re-traumatisation.*

RED FLAGS

RED FLAG	CLINICAL SIGNIFICANCE	ACTION IF PRESENT
Suicidal ideation or severe depression	FND has a high burden of psychological comorbidity and significantly elevated suicide risk in severe or long-standing cases	Involve mental health team immediately; follow safe messaging guidelines; do not proceed without psychiatric support
New or evolving neurological symptoms not previously assessed	Organic pathology can co-exist with FND — a new symptom should never be automatically attributed to FND without re-evaluation by neurology	Refer back to neurology urgently; document carefully; do not assume all new symptoms are functional
First-ever prolonged seizure or loss of consciousness	Functional seizures (PNES) need to be distinguished from epilepsy; first presentation requires medical evaluation and EEG	Ensure team has a management protocol; do not restrain during event; refer to neurology and epilepsy services
Disclosure of active abuse or safeguarding concern	Ongoing trauma can be the maintaining factor for FND and poses immediate risk to Chloe's safety	Follow mandatory safeguarding reporting protocol; involve social work; prioritise safety above therapeutic progress

PROBLEM LIST

PHYSICAL IMPAIRMENTS

- Left leg weakness — inconsistent, attention-modulated
- Left arm postural/action tremor — frequency entrainable
- Abnormal gait — left foot drag pattern
- Possible secondary deconditioning from activity avoidance
- Altered body-schema / limb awareness for left side

FUNCTIONAL ACTIVITY LIMITATIONS

- Difficulty walking safely and efficiently in the community
- Left arm tremor limits fine motor tasks and carrying objects
- Reduced independence in ADLs and community participation
- Potential impact on employment and social roles
- Fear-avoidance of movement maintaining disability cycle

PSYCHOSOCIAL / YELLOW FLAGS — VERY HIGH RELEVANCE IN FND

Chloe may hold unhelpful illness beliefs ("something structural is being missed"), catastrophise about her prognosis, have significant health anxiety, and demonstrate fear-avoidance of physical activity. Social support deficits, impact of symptoms on identity and roles (employment, relationships), and prior dismissive medical encounters are all clinically significant. These are not peripheral — they are core treatment targets alongside the motor impairments.

2 Pathophysiology of Working Diagnosis

WHAT IS HAPPENING NEUROLOGICALLY

> Functional Neurological Disorder is a disorder of neurological *function* — the brain is producing abnormal motor and sensory output despite structurally intact neural hardware, as confirmed by Chloe's normal MRI and nerve conduction studies. The current leading neuroscientific model is the
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> **PREDICTIVE PROCESSING FRAMEWORK**
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> : the brain is a Bayesian prediction machine that constantly generates prior predictions ("beliefs") about the body's movements and state, comparing them with incoming sensory signals to produce a "best guess" of what is happening. In FND, this system becomes dysregulated — the brain generates an overpowering, fixed prior that the limb cannot move normally, and this prediction overrides normal voluntary motor commands. Functionally, this manifests as symptoms that are real, involuntary, and neurologically generated, but not driven by structural damage to any specific pathway. Neuroimaging studies have shown altered connectivity between the prefrontal cortex (attention and prediction), the limbic system (particularly the amygdala — threat processing and emotional regulation), and the supplementary motor area (preparation of voluntary movement) — suggesting that heightened arousal and focused attention on the limb paradoxically suppress normal voluntary motor initiation. This is consistent with the key clinical feature: symptoms fluctuate with attention — they improve when Chloe is distracted and worsen when she focuses on them or feels observed. The precipitating psychological stress likely sensitised this prefrontal-limbic-motor circuit, functioning as a "trigger" event for the disorder.

WHY THIS PATIENT FITS — POSITIVE DIAGNOSTIC SIGNS OF FND

Positive Hoover Sign:
Left leg weakness on direct testing, but involuntary hip extension force returns when Chloe performs hip flexion on the right — this is the most specific bedside sign of functional leg weakness (near 100% specificity in research); it demonstrates that the neural circuitry is intact but the voluntary motor drive is blocked by the predictive system

Tremor entrainment:
Left arm tremor frequency changes when Chloe taps a rhythm with the unaffected hand — organic tremors are driven by fixed subcortical oscillators and maintain their own frequency regardless of competing tasks; entrainment is a validated positive sign of functional tremor

Distractibility of weakness:
Leg strength improves significantly when Chloe is distracted — consistent with an attention-mediated disorder of voluntary motor control rather than a fixed structural deficit

Normal MRI and nerve conduction studies:
Rules out structural lesion, inflammatory demyelination, and peripheral neuropathy — making an organic aetiological cause implausible as a primary explanation

Acute psychosocial precipitant:
Sudden onset following significant psychological stress is the classic trigger event; consistent with sensitisation of the predictive processing system

3 Differential Diagnoses

✦ Primary: Functional Neurological Disorder — Motor Subtype (FND)

Confirmed by positive clinical signs: Hoover sign, tremor entrainment, and distractibility — these are not signs of exclusion but validated positive findings with high specificity for FND. Modern neurology diagnoses FND on positive clinical evidence, not simply by ruling out alternatives. Normal MRI and NCS rule out structural and peripheral pathology. Sudden onset following psychosocial stress and the clinical profile (young woman, peak FND incidence 20–40s, female predominance 2–3:1) all fit.

Differential 1: Early Multiple Sclerosis (MS)

Supporting features initially: Young woman, unilateral motor symptoms, sudden onset — all consistent with a first MS relapse. However, MS would show periventricular or juxtacortical white matter lesions on T2/FLAIR MRI, abnormal visual evoked potentials, and possibly oligoclonal bands in CSF. The entirely normal MRI makes demyelinating disease very unlikely. The improvement in strength with distraction is also not a feature of UMN weakness from MS.

Differential 2: Essential Tremor

Supporting features initially: Young adult with action tremor, no structural cause. However, essential tremor is typically bilateral, symmetrical, and progressive — not acutely precipitated by psychological stress. Crucially, essential tremor does not entrain to externally imposed rhythms; its frequency is maintained by a fixed cerebello-thalamic oscillator. The unilateral tremor, acute onset, entrainment sign, and normal NCS make FND the far more parsimonious explanation.

WHY I FAVOUR FND

I favour FND over MS because the MRI is entirely normal — MS cannot produce motor symptoms without corresponding T2 lesions in the relevant pathways. I favour FND over essential tremor because the entrainment sign is present — this is impossible with a fixed organic tremor generator. Most importantly, I am making a positive diagnosis based on validated clinical signs, not simply a diagnosis of exclusion. This matters clinically: a positive FND diagnosis allows me to explain the diagnosis credibly to Chloe and begin targeted treatment.

4 Objective Assessment — Special Tests

SPECIAL TESTS

Test 1 — Hoover Sign

MEASURES Positive FND motor sign — tests whether lower limb weakness is voluntary (intentional motor drive absent) or associated with intact automatic motor output

HOW TO PERFORM

1. Patient lies supine. Explain you are going to test the strength in both legs.
2. Place one hand under the heel of the weak left leg. Ask Chloe to lift the left leg against your resistance — note the force produced (typically reduced in FND).
3. Now move your hand and ask Chloe to press DOWN with her right (unaffected) leg against resistance — hip extension task on the strong side.
4. While she presses the right leg down, feel for involuntary downward pressure (hip extension) returning in the left heel — without asking her to push it down.
5. Note: you can also check the reciprocal — ask her to push the right leg UP (hip flexion) and feel for involuntary hip flexion appearing in the left leg.

POSITIVE FINDING Involuntary extension force returns in the affected left limb during the contralateral task — demonstrates intact corticospinal circuitry for the movement; the deficit is in voluntary motor initiation (intentional drive), not in the structural pathway itself.

WHY I CHOSE IT The Hoover sign has the highest specificity of any bedside motor sign for FND. Documenting and being able to explain a positive Hoover sign allows me to make and communicate a confident, positive diagnosis — and importantly, to demonstrate to Chloe in real time that her motor circuitry is working, which supports the neurological explanation of FND.

Test 2 — Tremor Entrainment Test

MEASURES Distinguishes functional tremor (frequency entrains to external rhythm) from organic tremors (fixed frequency driven by a subcortical oscillator)

HOW TO PERFORM

1. Ask Chloe to hold the affected left arm outstretched so the tremor is visible. Observe and note the resting tremor frequency.
2. Ask her to tap a specific rhythm with her right (unaffected) hand — start with a slow, clear 2 Hz beat, counting aloud: "one, two, one, two..."
3. Observe whether the left arm tremor changes frequency toward the tapping rhythm, reduces in amplitude, or stops entirely during the cognitive demand of tapping.
4. Now change the tapping speed and observe the left arm response again.
5. Try a complex rhythm (irregular beats) — functional tremors often suppress entirely under this dual-task load.

POSITIVE FINDING (Functional) Left arm tremor changes frequency toward the tapping rhythm, or suppresses significantly during the task. Organic tremors (Parkinson's resting, essential, cerebellar) are generated by fixed subcortical circuits and maintain their own frequency regardless of competing motor tasks.

WHY I CHOSE IT Tremor entrainment is a validated, widely taught positive sign for functional tremor. It is also a powerful psychoeducation moment — I can show Chloe that her tremor is modifiable by where attention is directed, which supports the brain-network explanation I will give her about FND.

Test 3 — Gait Observation: Inconsistency Assessment

MEASURES Documents inconsistency in the gait impairment (a positive FND feature) and baselines the gait pattern as a treatment target

HOW TO PERFORM

1. Observe Chloe walking from one end of the room to the other with no instruction other than to walk normally — document foot clearance, step length, velocity, and left foot drag.
2. Ask her to walk while performing a cognitive dual task: "As you walk, count backwards from 100 in 7s."
3. Compare gait quality between conditions — note any improvement with the dual task.
4. Observe across multiple trials for intra-session variability — does the gait pattern change when she believes she is not being watched?
5. Note any features that do not match a consistent organic pattern (e.g., the drag appears on alternate trials, or changes sides briefly).

POSITIVE FINDING (Functional) Gait improves during dual-task or distraction; pattern is inconsistent across trials; features are atypical for a single organic pattern (e.g., not a pure UMN scissoring gait or LMN steppage gait).

WHY I CHOSE IT Gait is Chloe's primary functional complaint and the most visible, disabling symptom. Documenting it provides my clinical baseline, demonstrates the core FND feature of inconsistency for the examiner, and gives me direct measurable targets for gait retraining.

KEY NEUROLOGICAL OBSERVATIONS

Tone:
Often normal in FND; co-contraction (simultaneous agonist and antagonist firing) may create give-way or "collapsing" weakness pattern on manual muscle testing — distinct from true paresis

Reflexes:
Should be intact and symmetrical throughout — asymmetric or abnormal reflexes would prompt immediate reconsideration of an organic lesion

Sensory:
FND sensory symptoms often have a "split midline" or anatomically inconsistent glove-and-stocking pattern — note carefully but avoid over-investigating

Gait:
Observe foot drag, variability across trials, response to distraction; also test tandem walking — may be disproportionately impaired or paradoxically normal

Mental state:
Note affect, eye contact, engagement and willingness to participate — clinical markers of readiness for rehabilitation and psychological co-morbidity level that I communicate to the MDT

OUTCOME MEASURE

> **PRIMARY: PATIENT-SPECIFIC FUNCTIONAL SCALE (PSFS)**
>
> Chloe identifies her 3 most important functional goals and rates each 0–10 (0 = cannot perform, 10 = able to perform at pre-illness level). This is person-centred, highly responsive to meaningful change, and avoids the risk of using impairment-based measures that may paradoxically increase Chloe's focus on her symptoms. Minimum clinically important difference is 2 points.
>
> **SUPPLEMENTARY: PHQ-9 AND GAD-7**
>
> — given the very high prevalence of depression (PHQ-9) and anxiety (GAD-7) in FND, I screen at baseline and re-administer monthly. These scores are shared with the MDT to inform the psychology co-management pathway.

5 Treatment Plan

COMMUNICATING THE FND DIAGNOSIS — NON-STIGMATISING APPROACH

💬 HOW TO EXPLAIN FND TO CHLOE

- 💬 **Validate first, every time:** "Chloe, I want to be absolutely clear — your symptoms are real. The weakness in your leg and the tremor in your arm are not imagined, and you are not making them up or putting them on. I can see them and I believe you."
- 💬 **Use a credible, non-blaming analogy:** "Think of your brain as a computer with completely intact hardware — your MRI and nerve tests confirm that. But the software — the programmes the brain uses to send movement signals — is running incorrectly. This is what we call Functional Neurological Disorder: a problem with how the brain is working, not with its structure."
- 💬 **Reference the positive signs:** "I found some specific signs today that actually confirm this diagnosis. When I distracted you with another task, your leg got noticeably stronger. That tells us the pathways and wiring for movement are intact — your brain just isn't connecting them properly when your attention is focused on the leg."
- 💬 **Link the stress without causing blame:** "Significant stress can change the way the brain processes movement signals — similar to how stress causes migraines, or makes your heart race and your stomach churn. That's a brain-body link that is well established. It doesn't mean your symptoms are caused by stress in a simple way, and it absolutely doesn't mean this is your fault."
- 💬 **Offer genuine hope:** "FND is treatable. Research shows that physiotherapy specifically designed for FND — often combined with psychological support — can lead to significant and sometimes complete improvement. Many people do recover substantially."
- 💬 **Language to actively avoid:** Never say "there's nothing structurally wrong," "it's all in your head," "psychosomatic," "medically unexplained," or "conversion." These carry deep historical stigma and break therapeutic alliance immediately.

INITIAL PHASE — WEEKS 1–4 (2 SESSIONS/WEEK)

> **NEUROPHYSIO RATIONALE FOR FND**
>
> FND physiotherapy is a
>
> **BOTTOM-UP MOTOR RETRAINING APPROACH**
>
> . Rather than asking Chloe to "try harder" to move — which engages the very attentional mechanism maintaining the disorder — I use techniques that bypass voluntary motor focus: automatic movement initiation, rhythmic facilitation, distraction during movement, and positive reinforcement of normal movement when it appears. I also aim to shift Chloe's attentional relationship with her limb from threat-monitoring to confident automatic use. The therapeutic alliance is itself an active treatment component.

Exercise 1 — Rhythmic Auditory-Cued Walking (Automatic Gait Initiation)

What: Use a metronome or rhythmic music to initiate and regulate walking, shifting Chloe's attentional load from monitoring the affected leg to tracking an external rhythmic cue. Begin with bilateral stepping on the spot in standing (with support available), progressing to forward walking.

Why: Rhythmic auditory cueing engages the basal ganglia and supplementary motor area via an automatic, exogenous attentional pathway — distinct from the effortful voluntary pathway that is dysregulated in FND. This is why many FND patients walk better to music than on direct instruction. It also reduces the "monitoring" of the affected limb that maintains the functional deficit, consistent with the predictive processing model.

⬇ Regression	⬆ Progression
Marching on spot to metronome in standing, hands on parallel bars for safety. Therapist taps the rhythm on the patient's shoulder as an additional tactile cue. Focus entirely on the rhythm — do not instruct the patient to focus on the leg.	Forward walking to music with no support. Dual-task walking (conversation, counting). Variable environments (corridors, outdoors). Progressively remove the auditory cue and ask Chloe to internalise the rhythm. Progress to community walking.

Exercise 2 — Mirror Therapy / Limb Reintegration

What: Chloe places both arms into a mirror box, positioned so the reflection of the unaffected right arm appears as if it were the left arm moving normally. She performs slow, coordinated movements with the right arm while watching the reflected image. The brain receives congruent visual input of a normally moving left arm.

Why: Mirror therapy provides the brain with congruent motor-sensory feedback for the affected limb — it delivers the visual signal of normal movement to the body schema, updating the brain's distorted predictive model. It is also a top-down attentional redirection: Chloe is watching movement rather than monitoring a tremor. Good evidence base in FND and CRPS for limb reintegration and tremor reduction.

⬇ Regression	⬆ Progression
Simple slow unidirectional movements (forearm flexion/extension) while watching mirror image. Begin with 5-minute sessions — some patients initially find the visual incongruence unsettling; introduce gradually.	More complex bilateral movements. Functional task simulation (reaching for a cup, turning a door handle). Progressively reduce reliance on the mirror as normal movement patterns re-emerge and Chloe develops confidence in the limb without visual reassurance.

SAFETY CONSIDERATIONS

- Never confront, challenge, or test inconsistency in a way that Chloe perceives as disbelieving — this worsens FND by amplifying threat responses and attention toward the limb
- Frame positive clinical signs (Hoover, entrainment) as informative rather than accusatory: "These signs help us understand what type of problem this is, which is actually good news for treatment"
- Falls risk during gait retraining: sudden give-way can occur — use contact guarding and gait belt throughout; ensure clear, unobstructed space
- Monitor for symptom spread (new FND symptoms in other body regions) — common in early rehabilitation; document and discuss with MDT promptly without amplifying Chloe's anxiety about it
- Have a documented, team-agreed plan for dissociative episodes during sessions: remain calm, speak softly, do not restrain, position safely, allow the episode to resolve, and debrief gently afterward

PATIENT EDUCATION — KEY MESSAGES

"FND is not a psychiatric disorder — it is a neurological one, now recognised in the same classification systems as other neurological conditions. Your brain's networks are the issue, not your mind or your character."

"Because the brain learns and adapts, these circuit patterns can change. Rehabilitation is about retraining the networks — not pushing through the symptoms by force, but finding ways to help the brain rediscover how to move normally."

"Your symptoms will probably fluctuate — better some days, worse on others. This variability is actually characteristic of FND and is a positive sign that your brain's control is potentially changeable."

"Gradually increasing activity — with the right pacing — is important for recovery. Completely avoiding movement to protect yourself can maintain and entrench the pattern. We will build this together, carefully and at your pace."

PROGRESSION CRITERIA — MEASURABLE TARGETS

- ✓ PSFS score increases by ≥ 2 points on at least one identified functional goal
- ✓ 10-Metre Walk Test completed without foot drag on ≥ 3 of 5 consecutive trials
- ✓ Tremor absent or significantly suppressed for ≥ 5 continuous minutes during at least one session (documented)
- ✓ Chloe can self-identify ≥ 2 strategies that reliably improve her symptoms (demonstrating self-efficacy development)
- ✓ PHQ-9 and/or GAD-7 scores stable or showing improvement (confirming psychological co-management is effective)

LONG-TERM PLAN — WEEKS 4–12+

> MDT approach is the gold standard for FND and is now reflected in international clinical guidelines. I advocate for active collaboration with a clinical psychologist or psychiatrist skilled in FND — therapeutic modalities including CBT adapted for FND, Acceptance and Commitment Therapy (ACT), and trauma-informed care (if trauma is a maintaining factor) have the best evidence base. Occupational therapy input for work reintegration, pacing, and meaningful activity. Gradual supported return to valued roles — employment, social, recreational. Patient self-management education using validated public resources: *neurosymptoms.org* (developed by leading FND neurologist Professor Jon Stone, University of Edinburgh) and the *FND Hope* patient charity network. Regular physiotherapy review every 6–8 weeks as Chloe develops self-management skills, with the explicit goal of reducing dependence on formal therapy as recovery progresses.

🎓 Examiner Demonstration — Techniques Performable in the Room

1. **Tremor Entrainment Live Demonstration:**

   Ask Chloe to hold out her tremoring arm. Ask her to tap a rhythm with the other hand while you count at a different pace — narrate in real time what you are observing and why it is diagnostically significant and therapeutically informative. "Notice how the tremor is changing — this tells us the tremor is influenced by where attention is directed, which is exactly what we see in FND and it means we can work with it."

2. **Rhythmic Facilitation for Gait:**

   Stand beside Chloe, begin a clear verbal or clapping rhythm, and walk alongside her in step — narrate how the external auditory cue shifts her attentional demand away from the affected leg and facilitates more automatic, normal gait output. Point out the improvement in foot clearance that typically occurs. This directly demonstrates the core rehabilitation strategy.